

Multimodal Llms For Image-Based Fabric Composition Identification

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Abstract

With e-commerce shaping the way people buy clothes today, correctly determining what fabrics are made of is now a key requirement. Conventional methods are often constrained by hardware dependencies (e.g. NIR sensors), limited accessibility and poor fabric classification. We propose a parameter-efficient multimodal system that uses instruction-tuned LLMs to determine the proportion of fibers directly from fabric images. We compare two different multimodal models. The first is Meta's LLaMA 3.2-Vision 11B Instruct model, which builds on a separately trained image encoder and a vision adapter that integrates image-feature representations via cross-attention layers into a core language model. The second is Salesforce's BLIP2, which connects a ViT-G/14 image encoder to an OPT language model. In both the models, a light-weight regression head is appended to the final pooled representations and trained alongside LoRA injected adapters targeting the query, key and value projections within the transformer layers. The LoRA configuration is implemented using the PEFT (Parameter-Efficient Fine-Tuning), reducing trainable parameters to a few million for efficient adaptation. Experiments span six fiber categories -cotton, wool, polyester, linen, silk, and other fibers using a custom dataset of labelled textile images. The BLIP2-based model achieved a mean absolute error of 13.40 percentage points, outperforming the LLaMA 3.2 model, which recorded 14.87 percentage points on the test set. The evaluation results validate the use of vision-language models for estimating fiber composition and point toward their potential adoption in both consumer and industrial applications.

Keywords: Fabric Composition identification, Multimodal LLMs, Vision-Language Models, Regression, LoRA, PEFT, Textile Blends.

I. INTRODUCTION

Determining fabric composition is essential in the textile industry, influencing everything from quality assurance and eco-friendly manufacturing to online apparel markets. As e-commerce and resale markets continue to grow in the textile industry, reliable and interpretable fiber authentication solutions are now more critical than ever. They support everything from automated e-commerce product categorization and claim verification in online sales, to quality assurance in textile manufacturing and fiber sorting for recycling and reuse. Accurately identifying fiber content is especially crucial for textiles made for infants, medical patients, or people with sensitive skin. If synthetic fibers go undisclosed in garments marketed as pure, it can cause allergic responses, violate safety standards and harm a company's reputation.

Traditional methods often depend on equipment like NIR spectrometers, 3D touch sensors and microscope-based imaging systems. These methods do deliver good results,

but the expensive hardware and environment constraints make them unrealistic for large-scale or consumer deployment. Modern image-based models, including those built on CNNs or ViTs, generally require large amounts of labelled training data. Because they are designed to predict fixed fabric categories, they often struggle with fabrics that contain multiple fibers in varying proportions.

Our research puts forward a framework that uses BLIP and LLaMA 3.2 Vision, both multimodal LLMs, to interpret fabric images and estimate fiber proportions through instruction-based reasoning. Both the image and a short instruction are processed by the models. The user only supplies the fabric image, while the system internally attaches a fixed prompt to direct the inference process. The features produced by the model are then fed into a custom regression layer that estimates continuous percentage values for six fiber types: cotton, wool, polyester, linen, silk and other fibers category.

Instead of just classifying fabrics into one category, our method estimates the proportion of each fiber which makes it better for mixed materials. We attach a lightweight regression component to the vision-language model and finetune it alongside LoRA adapters, using mean squared error loss and dropout regularization strategy. We have evaluated the system using a dataset of 1,983 images. Our results indicate that the combination of instruction-tuned vision-language models with a regression head enables reliable and scalable fiber composition identification from everyday smartphone photos. This allows use of system in retail operations, monitoring of production and processes involving material reuse.

II. INDUSTRIAL RELEVANCE

This approach provides value in industrial settings as it allows identification of fiber content to occur at large scale using images from general sources and instruction in natural language form. Today, in the era of digital fashion, fabric content can be checked using just a phone or camera image avoiding the need for high cost tools like NIR or tactile sensors. Our system is built to serve both sellers and buyers. Sellers on e-commerce sites or in small studios can use the system to automatically label fabrics, avoiding the effort and expense of doing it by hand or sending them for lab testing. Online buyers can use the system to check if the product information is real, making them more confident and less likely to return items.

Allowing fabric composition to be checked from photos using multimodal models helps push the textile field towards more accessible, intelligent and easy-to-adopt AI technologies. Because it can understand natural language commands, the system can be used in chat assistants, mobile apps, and automated quality checks in retail.

Accurate fiber analysis becomes even more important in cases where textiles are intended for infants, healthcare environments or anyone prone to skin irritation. If clothes labelled as pure cotton actually contain synthetic fibers like polyester or rayon, they can cause allergies, skin irritation or make the fabric feel uncomfortable. Such mislabeling may also result in non-compliance with safety regulations. For example, Hungary's National Authority for Trade and Consumer Protection discovered that some products labelled '100% cotton' actually contained 40-60% polyester which was considered misleading to consumers[13]. This shows the importance of fiber content analysis to keep consumers safe, stay transparent and promote responsible production.

During quality inspection, fiber analysis provides means to confirm that production uses the right materials and that quality remains consistent across batches. Quality issues in production and fluctuations in supply materials can produce shifts in fiber ratios. These shifts affect the product's texture, lifespan and texture quality. For example, undiscovered

shifts in silk and cotton blend ratios can produce fabrics that do not satisfy the intended quality requirements. Adding fiber analysis using automated methods to the production process allows early identification of these issues. This approach helps maintain material quality and protects brand reputation, particularly for premium clothing lines. Sustainability efforts in textile industry rely heavily on accurate fiber detection to enable efficient recycling workflows. The workflow changes based on the fiber composition. Fabrics that are fully cotton can be recycled in closed-loop cycles, but blended textiles require more complex treatment. Circular fashion initiatives such as Renewcell[12], only accepts garments with at least 95% cotton content for conversion into Circulose, a biodegradable textile pulp. Automated fiber detection improves the sorting process, lowers the chance of mixing incompatible materials and encourages environment friendly textile workflows.

III. DATASET

The dataset for this study was developed in multiple stages to include different fabric types and maintain accuracy in fiber composition. It included detailed fabric images taken from popular fashion and textile sites such as H&M, Zara, J.Crew and Mood Fabrics. Since scraping tools were not reliable due to copyright and formatting problems, the data had to be collected manually from each site. During data cleaning, we came across some samples where composition values were inaccurate because the percentages did not sum to the expected 100%. We removed such inconsistent records to maintain good quality of data for this study. Each image was labelled with the exact composition which shows the percentage of cotton, wool, polyester, linen, silk and other fibers. This data was stored in a csv file which was indexed by image_id.

The dataset was arranged as a collection of image files in a directory accompanied by a CSV metadata file with per-image fiber composition labels. The dataset was randomly divided into three parts: training (80%), validation (10%), and testing (10%) using the `train\_test\_split` function from the scikit-learn library, ensuring that these proportions were consistent across runs. These divisions were used to train and evaluate a multimodal regression model. The architecture was fine-tuned using a custom regression head. Using a partially frozen vision-language model as a base, we can make predictions. Image-text pairs provide a way to understand the different parts of fiber compositions.

Fig. 1 Representative samples (Img 01–Img 06, arranged left-to-right and top-to-bottom) from the fiber composition dataset, illustrating variations in texture and color across fiber types.





TABLE 1 Composition details(in %) of the sample images shown in Figure 1.

Img_id	Cotton	Wool	Polyester	Linen	Silk	Other fibers
Img_01				100		
Img_02					95	5
Img_03	14	29	15			40
Img_04			100			
Img_05				51	49	
Img_06		97				3

IV. RELATED WORK

Analysis of fabric composition went through different stages of development. Manual analysis approaches were used to create the first generation of feature descriptors of fabric patterns. Subsequently, computerized approaches, with the application of deep learning methods, replaced these approaches. Texture properties, as well as the surface properties of fabric, were obtained using the application of robust approaches (GLCM, Gabor filters, and LBP approaches). The approaches proved successful, especially in the laboratory setting, however, their performance decreased dramatically, especially during the analysis of real world images due to unpredictable lighting, texture variation, and poor image quality.

The use of the Convolutional Neural Network (CNN) in fabric categorization presented an improvement over the traditional means used to tackle similar problems. The use of hierarchical features by the CNN made the models better equipped to deal with the complexities involved in the texture and pattern of materials. As shown by previous studies [7], the use of Ensemble CNN architectures is capable of increasing the accuracy level in multi-class fabric classification problems. The models are usually computationally intensive and are designed for the classification of discrete fabric categories and not capable of estimating the compositions of materials blended together. It was recognized

that the above mentioned architectures had severe limitations, hence new architectures with reduced complexities relative to the above ones have been developed.

These new models are called Texture-Convolutional Neural Networks (T-CNNs) [6] and wavelet scattering networks (ScatNet) [6]. These new models possess smaller sizes in their models but also mitigate better upon translation invariance in respect to the previous models base architecture designs. Other scientists pursued a further strategy by proposing models that are a combination of two models that contain the greatest elements from each strategy. A prominent model that practices this strategy includes NmzNet [8] and their way of capturing traditional models such as Gabor Filtering and Local Binary Patterns (LBP), combined with recent models from deep learning strategies that are designed in a way that can capitalize on a variety of supplementary representations that will enhance their discriminative capabilities between different fabrics.

Apart from these image-based solutions, a number of other sensory solutions have also been researched, including near-infrared spectroscopy, tactile imaging, and optical coherence tomography [1]. Although these can also prove quite effective with a high degree of accuracy, they do require specific hardware, making them rather impractical from a consumer or low-resource perspective. Recent years have also seen the advent of vision language models on a large scale and multimodal learning as a new means of image analysis. Instruction-tuned models include examples like BLIP, LLaMA Vision and LLaVA, all of these having shown a strong degree of generalizability through combining both visual and text information into an integrated process. The effectiveness of these models in making predictions of the compositions of textiles as a continuum has also not been well-explored.

V. PROPOSED METHOD

In an effort to move beyond the shortcomings of traditional methods based on classification in the analysis of fabric composition, the proposed study considers the multimodal regression framework of an instruction-tuned vision-language model that uses both image data of fabrics and text prompts to predict the proportion of fibers within fabrics.

A. Multimodal Input Representation

The model has two input components: a fabric image and a constant natural language instruction: "What is the fabric composition of this textile? Please analyze the fibers present." The instruction assists in directing the model's attention to the regression task. The two components are subsequently processed together as a text and image in a processor that belongs to the model and is adept at embedding them in a common feature space that assists in determining the percentage composition of fibers.

B. Vision-Language Backbone: LLaMA 3.2 Vision

The primary architecture utilized in this study is Meta's LLaMA 3.2 11B Vision-Instruct model - a large-scale multimodal transformer that integrates both vision and language understanding within a unified framework. The model leverages a SigLIP-based Vision Transformer (SigLIP-L/14) as its visual encoder, which operates on 14×14 image patches and is trained using a sigmoid contrastive loss to enhance image-text alignment. This configuration allows the system to extract rich visual representations and seamlessly fuse them with linguistic context.

Prior to model input, all images were preprocessed using the AutoProcessor associated with the LLaMA 3.2 11B Vision model. Each image was resized to 560×560 pixels, a higher- resolution input compared to standard vision transformers, to preserve fine grained detail. The processor also handled color space standardization by converting all images to RGB format and normalizing pixel values according to model-specific statistics derived from the LLaMA 3.2 Vision encoder’s pretraining data. Consistent with CLIP-style normalization, channel-wise mean and standard deviation values were applied, and the model’s multi resolution support enabled adaptive image scaling for improved visual fidelity.

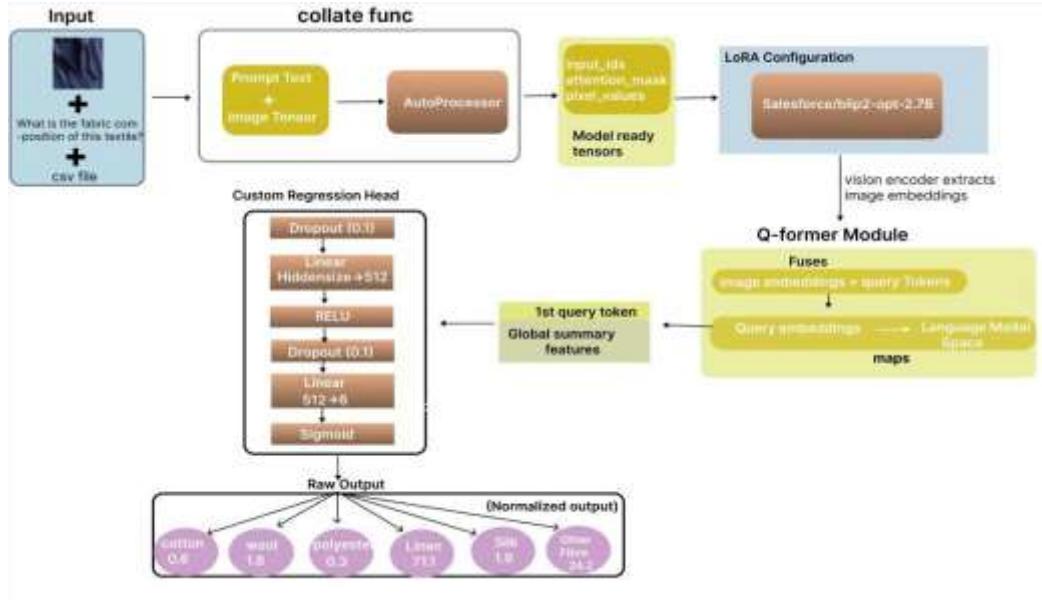
For textual input, prompts were structured and tokenized using the LLaMA 3.2 Tokenizer, which follows the model’s chat- template format based on role-specific conversation structure (e.g., user and assistant). The tokenizer employs an extended SentencePiece implementation with Byte-Pair Encoding (BPE) and automatically formats text using the `apply_chat_template` method to ensure consistency with LLaMA’s conversational fine-tuning.

During inference, image and text streams are processed in parallel through separate encoder modules. Cross-attention mechanisms are then employed to fuse visual and textual embeddings, aligning them in a shared latent representation space. A vision-language projection layer adjusts the dimensionality of visual features to match the language model’s hidden size of 4096, facilitating smooth integration within the transformer backbone. The resulting pooled multimodal embeddings encapsulate both semantic and visual context, forming the high-level representations used for downstream prediction tasks such as fiber composition classification.

C. Vision-Language Backbone: BLIP2

This study employs Salesforce’s BLIP-2 (Bootstrapped Language-Image Pretraining) framework as an alternative vision-language architecture. The BLIP-2 system combines a ViT-G/14 vision encoder with an OPT-2.7B language model, enabling efficient multimodal understanding. Before feeding data into the model, all images were standardized using the pretrained BLIP-2 processor (Blip2Processor). Each image was automatically resized to 224×224 pixels, converted to RGB format, and normalized according to ImageNet mean and standard deviation values. The processed images were then transformed into tensor representations compatible with the BLIP 2 vision encoder. For the text component, prompts were tokenized with the GPT2Tokenizer (optimized for OPT models) using Byte-Pair Encoding (BPE). Special tokens marking the beginning and end of sequences were inserted automatically, and all sequences were padded and masked to maintain uniform length (up to 512 tokens) across batches. Both the visual and textual inputs were processed in parallel through the BLIP-2 processor to ensure synchronized alignment. Within the Q-Former module, the image embeddings and text tokens are jointly encoded to form a unified representation. The ViT-G/14 encoder extracts dense visual features from 14×14 image patches, which are then fused with the textual features through a lightweight transformer decoder, producing pooled multimodal embeddings that capture rich cross-modal relationships. These embeddings serve as the input for subsequent downstream prediction tasks.

Fig .2 : Multimodal architecture for textile composition estimation using LoRA adapted BLIP2 and a custom regression head



D. Feature Extraction Approaches

❖ BLIP

Vision Encoder for Image Feature Extraction

The vision encoder extracts visual features from the input image and converts them into a sequence of visual embeddings. This process can be mathematically expressed as:

$$I = \text{VisionEncoder}(X_{\text{image}}) \quad (1)$$

where,

- X_{image} is the input image tensor (pixel values).
- $I \in R^{B \times N \times D_v}$ is the sequence of visual embeddings.
- N = number of visual tokens (image patches).
- D_v = is the dimensionality of the visual embedding.

The extracted visual embeddings serve as high-level representations of the image content and are subsequently utilized for multimodal alignment and downstream tasks.

1. Query Tokens Attend to Image Embeddings via Q-Former

$$Q_{\text{out}} = Q\text{Former}(Q_{\text{init}}, I) \quad (2)$$

where,

- $Q_{\text{init}} \in R^{B \times M \times D_q}$ are learned query tokens.
- $Q_{\text{out}} \in R^{B \times M \times D_q}$ are query outputs after cross-attention to L.
- M = number of query tokens.
- D_q = query embedding dimension.
- Q-former is a transformer that allows query tokens to attend over vision tokens, producing fused multimodal features.

2. Query output is projected to language model space

$$L = Q_{out} \cdot W_{proj} \quad (3)$$

where :

- $L \in R^{B \times M \times D_t}$ is the language space projected feature.
- $W_{proj} \in R^{D_q \times D_l}$ is the projection matrix.
- D_l : hidden size of language model (form BLIP2 OPT backbone).

3. First query Token Representation is used for Regression

$$f = L_{[:,0,:]} \quad (4)$$

We pick the first token embedding from L, shape $R^{B \times D_t}$, to represent the image as it captures the global summary features.

4. Pass through Regression Head

$$\widehat{y} = \sigma (W_2 \cdot ReLU (W_1 \cdot f + b_1) + b_2) \quad (5)$$

where:

- $\widehat{y} \in R^{B \times 6}$ is the predicted fiber composition percentages.
- σ = Sigmoid activation ensures [0,1] bounded output.
- $W_1 \in R^{512 \times D}$, $b_1 \in R^{512}$ - first linear layer
- $W_2 \in R^{6 \times 512}$, $b_2 \in R^6$ - second linear layer.

❖ LLaMA

1. The LLaMA Vision + Language model processes the tokenized multimodal input (images + prompt).
2. Extract the last hidden states from the language model. These are token-level representations of the multimodal input.

$$\begin{aligned} H \\ = LLaMA(input_tokens) \end{aligned} \quad (6)$$

- $H \in R^{B \times T \times D}$

- B : batch size
- T : total sequence length (depends on image + prompt tokens)
- D : hidden size (e.g., 4096)

H is the last layer of hidden state output for all tokens.

3. Instead of using a single token, we take the mean across sequence length and simple average pooling.

$$f = \frac{1}{T} \sum_{i=1}^T H_{i,j;k} \quad (7)$$

where, $f \in \mathbb{R}^{B \times D}$

This pooled vector represents a global summary of the multimodal input.

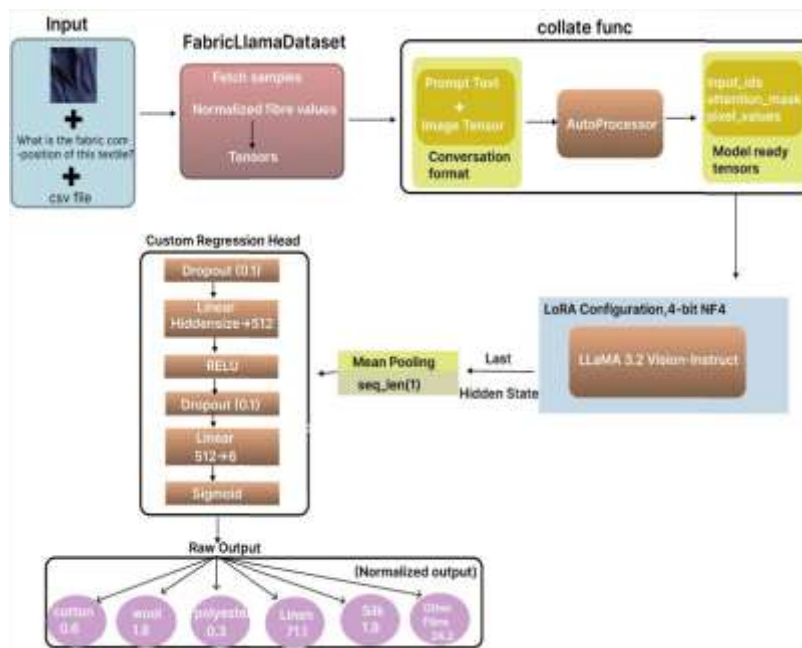
4. Pooled vector is sent to the regression head to predict fiber percentages.

$$\hat{y} = \sigma (W_2 \cdot \text{ReLU} (W_1 \cdot f + b_1) + b_2) \quad (8)$$

where:

- $W_1 \in \mathbb{R}^{512 \times D}$, $b_1 \in \mathbb{R}^{512}$ - first linear
- $W_2 \in \mathbb{R}^{6 \times 512}$, $b_2 \in \mathbb{R}^6$ - second linear layer
- $\hat{y} \in \mathbb{R}^{B \times 6}$ - predicted fiber percentages bounded [0,1]
- σ - sigmoid activation to bound outputs between 0 and 1

Fig. 3: Architecture of proposed FabricLlama framework for textile composition Vision-Instruct and a custom regression head



The BLIP2 and LLaMA models differ in how visual and multimodal features are extracted and passed to the downstream regression head. The BLIP2 architecture incorporates a dedicated Q-Former module, which learns a set of query tokens designed to extract high-level semantic information from the visual encoder. These query tokens are projected into the language model's embedding space and serve as compact, task-relevant representations for vision-language alignment. This targeted mechanism is optimized for tasks requiring precise cross-modal understanding, such as image-text reasoning or grounded regression. Conversely, the LLaMA 3.2 Vision-Instruct model relies on a generic mean pooling strategy. After jointly encoding image-text inputs within a unified transformer backbone, the hidden states of the language model are averaged across the sequence dimension to produce a fixed-size vector. While this approach is model-agnostic and computationally lightweight, it may dilute salient token-level features critical for fine-grained regression tasks. These differences reflect broader design philosophies: BLIP2 emphasizes modularity and specialized feature distillation (via Q-Former), while LLaMA leverages model scale and shared attention mechanisms to learn relevant features implicitly.

E. Regression Head Architecture

The pooled multimodal embeddings generated by the vision-language backbone are passed through a lightweight multi-layer perceptron (MLP) regression head, which is trained jointly with LoRA adapters applied to the transformer layers. This results in a hybrid setup with partial fine-tuning. The MLP maps the high-dimensional representations to six fiber composition percentages: cotton, wool, polyester, linen, silk, and other fibers. The head comprises:

- A dropout layer ($p = 0.1$) for regularization
- Two fully connected linear layers with ReLU activation,
- A final sigmoid activation to constrain outputs to the $[0, 1]$ interval.

Sigmoid Activation Function:

$$\sigma(x) = \frac{1}{1+e^{-x}} \quad (9)$$

This six-dimensional output vector represents the predicted fiber proportions. Post-processing normalization ensures that the outputs sum to 100%, enabling interpretable and consistent composition estimates.

F. Training Strategy and Optimization

The regression head and LoRA-adapted parameters are optimized during training, while the rest of the backbone weights remain frozen. A standard mean squared error (MSE) loss function is used to improve generalization and stability. The AdamW optimizer is employed, along with a ReduceLROnPlateau scheduler that adjusts the learning rate based on validation loss. Early stopping is used to prevent overfitting. To enhance computational efficiency during training on CUDA-enabled devices (NVIDIA GeForce RTX 4090), mixed-precision training is employed using PyTorch's autocast and GradScaler modules. Additionally, gradient clipping (`clip_grad_norm_`) is applied to limit the norm of gradients, improving stability during backpropagation and preventing gradient explosion in the transformer layers.

G. Model Precision Strategies

A critical distinction between the BLIP2 and LLaMA variants lies in their numerical precision configurations during training and inference. The BLIP2 model was fine-tuned in full 32-bit floating-point (float32) precision. This ensures numerical stability and avoids quantization artifacts, preserving gradient fidelity throughout the training process. However, it comes at the cost of increased memory consumption, requiring approximately 4 bytes per parameter and limiting the scalability of larger models. In contrast, the LLaMA 3.2 Vision-Instruct model was deployed using 4-bit quantization based on the NF4 (Normalized Float 4) format, implemented via the BitsAndBytes library. This quantization reduces memory usage to approximately 0.5 bytes per parameter, enabling the deployment of the 11B model on consumer-grade hardware. While quantization introduces minor loss of numerical precision, the reduced resource footprint allows for the use of significantly larger models. This trade-off between computational efficiency and slight degradation in accuracy underpins most architectural and implementation differences.

H. Evaluation Metrics

Model performance is tested using four standard regression metrics are used to evaluate how well a regression model works. These metrics help us understand how accurately the model predicts outcomes. They are important for assessing the model's performance and for comparing different models. Common regression metrics include the mean squared error (MSE), the root mean squared error (RMSE), the mean absolute error (MAE), and the coefficient of determination (R^2). Each of these metrics provides a different way to measure the model's accuracy, and they may be used together to get a complete picture of how well the model works.

● **Mean Absolute Error (MAE):** Measures average absolute deviation.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (10)$$

● **Root Mean Squared Error (RMSE):** Penalizes larger errors more heavily.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (11)$$

Tolerance Accuracy: Defined as the percentage of predictions within $\pm 10\%$ of the true fiber proportion per class.

$$Tolerance\ Accuracy = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \leq 0.10 \times 100\% \quad (12)$$

All predicted outputs are clamped to the $[0, 1]$ range using a sigmoid activation and are normalized afterward to ensure they form valid fiber percentage distributions summing

to 100%.

RESULTS

A. Training and Validation

We evaluated two multimodal architectures BLIP2 OPT (float32 precision) and LLaMA 3.2 Vision-Instruct (4-bit quantized) for the task of fiber composition regression. In both models, the core vision-language backbone (OPT in BLIP2, and LLaMA-3.2) remained frozen during training, only the LoRA adapters and a lightweight regression head were fine-tuned. Training was performed using the Mean Squared Error (MSE) loss function and the AdamW optimizer, with an initial learning rate of $1e-4$.

A ReduceLROnPlateau learning rate scheduler was utilized to adjust the learning rate. The learning rate should be adaptively decreased when the validation loss is observed. The model's performance plateaued, and early stopping was used to prevent overfitting.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (13)$$

where:

- y_i = Ground truth fiber percentage for class
- $\hat{y}_i$ = Predicted fiber percentage
- n = Number of fiber types (6 in this case)

Smooth MSE vs no. of epochs for BLIP2 Smooth MSE: computes Mean Squared Error between predictions and slightly smoothed targets to improve stability and prevent overfitting.

$$L_{SmoothMSE} = \frac{1}{n} \sum_{i=1}^N \left(\hat{y}_i - \left[y_i(1-s) + \frac{s}{2} \right] \right)^2 \quad (14)$$

where:

- $\hat{y}_i$ be the prediction for the i^{th} fiber
- y_i be the original target
- S be the smoothing factor (e.g., 0.01)

For BLIP2, full-precision training enabled stable convergence across epochs. The model's Q-Former module played a crucial role by sending a set of learned queries to the visual encoder and extracting task-specific features from the image. This component ensured that only the most relevant information such as fiber textures or fabric patterns was passed on to the text decoder. The design emphasized modularity and task awareness, helping BLIP2 achieve low validation losses and strong generalization.

On the other hand, LLaMA 3.2 Vision-Instruct employed a simpler feature aggregation mechanism. Visual features encoded into language tokens were pooled using a mean pooling strategy across the final hidden states. While this method is computationally light, it can average out important localized details. Additionally, the

LLaMA model was trained for fewer epochs, limiting the opportunity for the scheduler to adjust the learning rate effectively. Combined with the use of 4-bit quantization, which introduces minor numerical imprecision, these factors led to its relatively lower performance despite the advantage of being memory-efficient and capable of running large models on modest hardware.

TABLE 2: Validation Loss Summary

Epoch	BLIP-2 Val RMSE (%)	BLIP-2 Val Raw MSE	LLaMA Val RMSE (%)	LLaMA Val Raw MSE
1	26.62	0.070869	25.84	0.066747
2	26.02	0.067702	25.85	0.066804
3	26.05	0.067849	25.61	0.065585
4	25.93	0.067237	25.37	0.064387
Best (Val)	25.93	0.067237	25.37	0.064387

B. Test Set Evaluation

The final model's performance was evaluated using a separate test set. The performance metrics are summarized below.

TABLE 3: Performance Metrics

Metric	BLIP2	LLaMa
Mean Absolute Error(MAE)	13.40%	14.87%
Root Mean Square Error (RMSE)	25.21%	25.37%
Raw MSE	0.06227	0.06435

Tolerance Accuracy ($\pm 10\%$)	69.24%	63.60%
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Per-Fiber Evaluation

The model's prediction accuracy was analyzed per fiber class. The results are given in Table 4.

TABLE 4: Per-Fiber Performance

Fiber Type	BLIP2 MAE (%)	LLaMA MAE (%)
Cotton	25.72	30.36
Wool	11.37	14.15
Polyester	22.80	24.14
Linen	11.99	12.17
Silk	6.26	6.43
Other Fibers	2.26	1.98

C. Normalization Strategies

To ensure interpretability, predicted percentages were normalized using multiple strategies. Sum normalization and top-k normalization retained numerical fidelity. Softmax-based methods yielded overconfident single-class outputs which were suitable for pure fabrics. Hence, sum normalization was adopted, as it performed better in most cases and provided a more reliable normalization strategy for generalization across blended fabrics.

Softmax Normalization: Converts logits into a probability distribution over K classes.

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \text{ for } i = 1, 2, \dots, K \quad (15)$$

Sum Normalization: Scales a vector so that the sum of its elements is 1.

$$x_i^{\text{normalized}} = \frac{x_i}{\sum_{j=1}^N x_j} \quad (16)$$

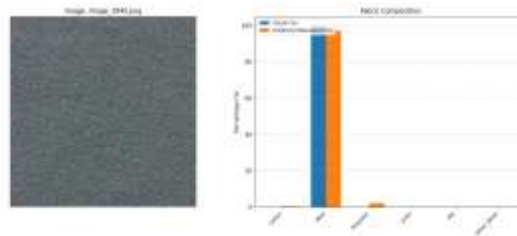
E. Sample Predictions

Sample predictions comparing ground truth vs predicted outputs (normalized) are presented in Table III and Table IV for selected test samples. All predictions are rounded to one decimal.

TABLE 5: Sample predictions (normalized) - BLIP2 C=Cotton, W=Wool, P=Polyester, L=Linen, S=Silk, O = Other

Sample	GT Composition (C,W,L,S,O)	BLIP2 Predicted Composition (C,W,P,L,S,O)	MAE (%)
1	[99.0, 0.0, 0.0, 0.0, 0.0, 1.0]	[94.7, 0.6, 3.0, 1.7, 0.4, 0.6]	1.73
2	[59.0, 0.0, 0.0, 0.0, 0.0, 41.0]	[94.6, 0.6, 2.4, 2.4, 0.6, 0.6]	13.65
3	[100.0, 0.0, 0.0, 0.0, 0.0, 0.0]	[95.6, 0.3, 2.4, 1.6, 0.3, 0.4]	1.56
4	[0.0, 0.0, 0.0, 100.0, 0.0, 0.0]	[19.7, 3.2, 3.4, 61.5, 2.6, 2.2]	11.60
5	[0.0, 53.0, 47.0, 0.0, 0.0, 0.0]	[13.7, 44.6, 37.5, 3.9, 4.9, 2.3]	7.11

Fig .4: Input fabric image (left) with actual vs. predicted fiber composition percentages (right) for test samples



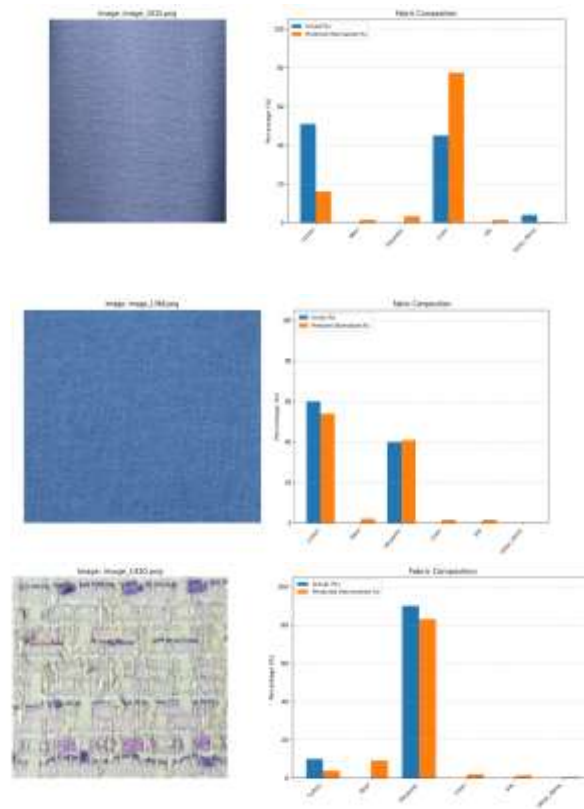


Fig .5 : Predicted fabric composition versus ground truth using LLaMA

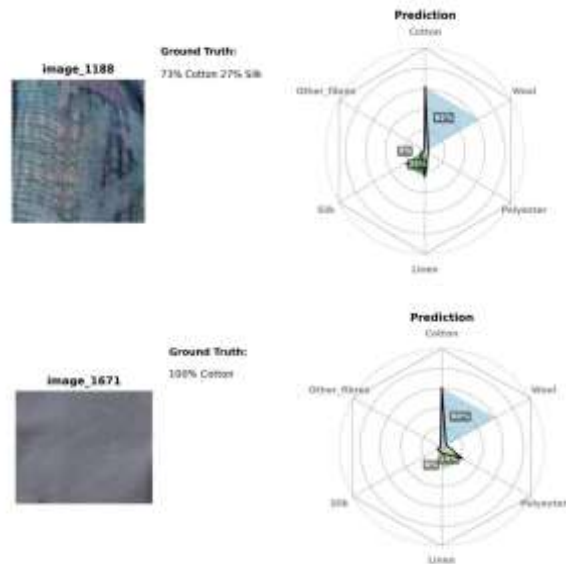


TABLE 6: Sample predictions (normalized) - BLIP2 C = Cotton, W = Wool, P = Polyester, L = Linen, S = Silk, O = Other

Sample	GT Composition (C,W,P,L,S,O)	LLaMA Prediction	MAE (%)
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1	[99.0, 0.0, 0.0, 0.0, 0.0, 1.0]	[75.0, 3.5, 14.2, 4.7, 1.6, 1.1]	8.00
2	[59.0, 0.0, 0.0, 0.0, 0.0, 41.0]	[72.1, 4.5, 14.7, 3.8, 4.0, 1.0]	13.00
3	[100.0, 0.0, 0.0, 0.0, 0.0, 0.0]	[83.5, 0.7, 11.7, 2.1, 1.5, 0.5]	5.52
4	[0.0, 0.0, 0.0, 100.0, 0.0, 0.0]	[14.1, 0.7, 1.0, 82.8, 1.1, 0.2]	5.73
5	[0.0, 53.0, 47.0, 0.0, 0.0, 0.0]	[16.2, 40.9, 32.2, 6.3, 3.6, 0.9]	8.99

Fig .6: Input fabric image (left) with actual vs. predicted fiber composition percentages (right) for test samples using BLIP2

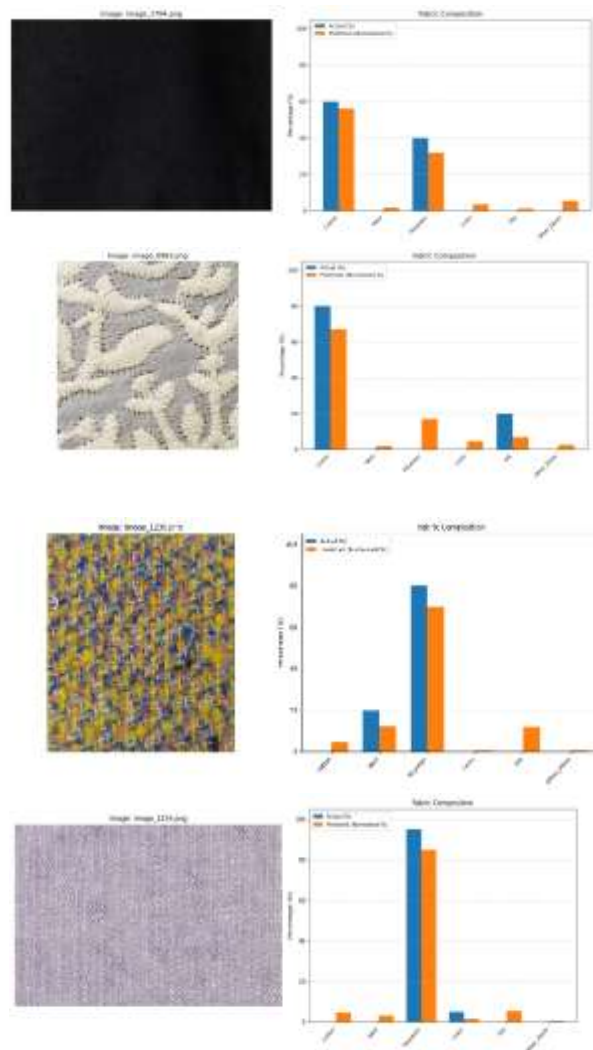
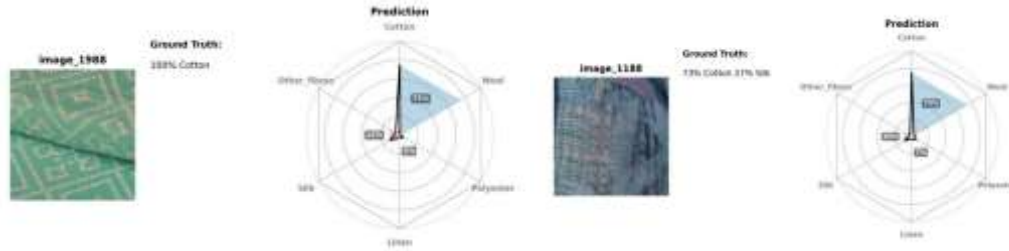


Fig. 7 : Predicted fabric composition versus ground truth using BLIP2

VII. CONCLUSION

We investigated two vision-language models for the task of predicting fabric composition using a regression head which is the focus of this study. The BLIP2 OPT model was trained with full 32-bit precision and incorporated a dedicated Q-Former to extract focused vision features relevant to the regression prompt. In contrast, the LLaMA 3.2 Vision-Instruct model operated in a 4-bit quantized mode and used a mean pooling strategy to summarize the hidden states from its language model. Both models utilized LoRA adapters to fine-tune only a small number of parameters and shared the same regression head design and training configuration.

Both models were trained for four epochs as additional training led to noticeable overfitting. This effect was mainly caused by the imbalanced dataset where most of the samples were made up mostly of cotton and polyester. While this inequality limited the models capacity to generalize to less prevalent fiber types. Reflecting real-world developments in the fabric industry, where mixtures with a significant amount of cotton, polyester, or a combination of both, these pairings are the most common. These findings underline the value of having diverse datasets in multimodal learning is significant. To boost performance, one could use augmentation or weighted sampling. While the overall results were similar, BLIP2 handled new samples a little better than LLaMA. This suggests that its architecture may be more effective at capturing cross-modal relationships in the presence of data imbalance.

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